

Hydrodynamics of Large Language Models

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By mapping the discrete layer depth to a continuous time variable, the processing of tokens within deep self-attention networks can be modeled as a continuous dynamical system. Defining an equivalent Knudsen number based on the mean inter-token distance, the high token density within active semantic regions permits the application of the continuum hypothesis. Operating in a collisionless, mean-field regime, as tokens interact via a non-local potential rather than kinetic scattering, the continuum limit of the Heavy Ball Transformer reveals that the latent space dynamics obey pressureless, compressible Navier-Stokes fluid equations. The stochasticity of algorithmic Dropout maps to microscopic Brownian diffusion governed by a Fokker-Planck continuity equation, while the Heavy Ball inertia yields an independent macroscopic momentum equation. Lacking a thermodynamic pressure analog, the non-local semantic attraction of the attention mechanism inevitably drives a degenerate spatial singularity. Through a Cauchy-Stokes decomposition of the macroscopic specific stiffness tensor, we demonstrate that this architectural rank collapse is fundamentally homologous to a hydrodynamic Jeans-like instability in a self-gravitating dust cloud. To natively arrest this collapse, we propose a structured operator that decouples the d -dimensional latent space into $d/2$ independent 2D planes, locking the fluid into an orbital steady-state of Solid-Body Rotation (SBR). This shear-free geometry prevents spatial collapse, recasting semantic contextualization as a physical process of angular phase synchronization governed by the Kuramoto order parameter, while the optimizer dynamically translates and reorients these semantic accretion disks to cluster correlated concepts.

Introduction and the Latent Continuum.—Large Language Models (LLMs) process sequences of tokens through stacked layers of multi-head self-attention [1]. The geometric rank collapse of latent representations in these deep networks is a known architectural vulnerability [2]. To investigate the macroscopic dynamics of this collapse, we model the discrete depth of the deep neural network as a continuous dynamical system [3].

We consider a context window of N tokens operating within a latent semantic manifold of characteristic length L and intrinsic dimension d , where d is significantly smaller than the ambient embedding space [4]. We define the mean inter-token distance as $\lambda \sim L/N^{1/d}$. While the global Knudsen ratio (λ/L) may be order unity due to the curse of dimensionality, tokens cluster heavily in active semantic regions where local density is high. Crucially, the attention mechanism constitutes a non-local interaction potential rather than hard-sphere kinetic scattering. Therefore, the system operates in a collisionless, mean-field regime analogous to a Vlasov plasma or self-gravitating cold fluid [5]. This locally dense, mean-field limit justifies the transition to continuous Eulerian fields: a compressible fluid density $\rho(\mathbf{x}, t)$ and a macroscopic velocity field $\mathbf{v}(\mathbf{x}, t)$. Furthermore, the absence of localized momentum-scattering collisions precludes local thermal equilibrium, fundamentally stripping the resulting macroscopic momentum equation of a thermodynamic pressure analog [6].

To establish a rigorous hydrodynamic framework, we first define an explicit conceptual mapping between the computational architecture of LLMs and the macroscopic variables of fluid mechanics. We treat the discrete seman-

tic representations (tokens) as individual fluid parcels, and the continuous network depth as the temporal evolution of the system. In this physical analogue, the self-attention mechanism acts as a non-local, mean-field interaction potential, conceptually akin to self-gravity or Vlasov-type electromagnetic coupling. While standard Transformers operate in an overdamped, kinematic regime, the introduction of Heavy Ball momentum [7, 8] explicitly grants these tokens macroscopic inertia, permitting the definition of a true momentum equation. Finally, stochastic regularization techniques natively employed in deep learning (e.g., Dropout [9]) are mapped to microscopic thermal fluctuations, driving macroscopic Brownian diffusion.

The Heavy Ball Navier-Stokes Fluid.—Mathematically formalizing this framework, we utilize the Heavy Ball (HB) Transformer architecture [7, 8] to mitigate the aforementioned rank collapse. Incorporating standard Dropout regularization [9] as the source of discrete stochastic noise ξ_l , the layer-to-layer update of a token's coordinate $\mathbf{X}$ is given by:

$$\mathbf{X}_{l+1} = \mathbf{X}_l + \alpha(\mathbf{X}_l - \mathbf{X}_{l-1}) + \mathcal{A}(\mathbf{X}_l) + \xi_l. \quad (1)$$

By converting the discrete layer index l to a continuous time variable t (where $\Delta z \equiv \Delta t \rightarrow dt$), defining the effective damping $\gamma = (1 - \alpha)/\Delta t$ (which requires $\alpha \rightarrow 1$ in the continuum limit), and scaling the discrete attention mechanism to a macroscopic force $\mathbf{F}_{att} \equiv \mathcal{A}/\Delta t^2$ [see Supplemental Material (SM) Sec. S1 for the proper well-posed limit scaling], the network converges to a damped, driven Langevin equation:

$$\ddot{\mathbf{X}}_t + \gamma \dot{\mathbf{X}}_t = \mathbf{F}_{att}(\mathbf{X}_t) + \sqrt{2\kappa} \dot{\mathbf{B}}_t, \quad (2)$$

where $\mathbf{B}_t$ is a standard Brownian motion and the macroscopic diffusion coefficient κ corresponds to the scaled variance of the Dropout. To transition from tracking discrete token trajectories to a continuous Eulerian framework, we define the local empirical fluid density as $\rho(\mathbf{x}, t) = \frac{1}{N} \sum_{j=1}^N \delta(\mathbf{x} - \mathbf{X}_j(t))$. Assuming unit latent mass ($m = 1$), the macroscopic deterministic force $\mathbf{F}_{att}$ is governed by a scalar Softmax interaction kernel $\mathcal{K}(\mathbf{x}, \mathbf{y})$ and the network’s linear weight matrix $\mathbf{W}$. In the mean-field limit ($N \rightarrow \infty$), this converges to a non-local spatial integral over the density measure [see SM Sec. S2]:

$$\mathbf{F}_{att}(\mathbf{x}, t) = \int \mathcal{K}(\mathbf{x}, \mathbf{y}) \mathbf{W}(\mathbf{y} - \mathbf{x}) \rho(\mathbf{y}, t) d\mathbf{y}. \quad (3)$$

Because the discrete spatial update is scaled by the squared time increment to yield a macroscopic acceleration for unit-mass tokens, the linear weight matrix $\mathbf{W}$ acquires units of inverse time squared ($1/s^2$). By operating directly on the spatial displacement ($\mathbf{y} - \mathbf{x}$) within the integral, $\mathbf{W}$ functions physically as a macroscopic specific stiffness-like tensor. The Softmax kernel $\mathcal{K}(\mathbf{x}, \mathbf{y})$ acts as a scalar weight, modulating the magnitude of this linear interaction based on the localized semantic similarity between fluid parcels.

Macroscopically, the evolution of the token density dictated by this stochastic dynamic is governed by the compressible Fokker-Planck continuity equation [see SM Sec. S3 for the formal Itô derivation]:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = \kappa \nabla^2 \rho. \quad (4)$$

Because the Heavy Ball formulation possesses intrinsic inertia, we transition directly to the Eulerian frame where the particle acceleration maps to the material derivative $D\mathbf{v}/Dt$. This maps the system to an independent continuous Navier-Stokes-like momentum equation [see SM Sec. S4 for the formal Lagrangian-to-Eulerian derivation]:

$$\frac{D\mathbf{v}}{Dt} = -\gamma \mathbf{v} + \mathbf{F}_{att}(\mathbf{x}, t). \quad (5)$$

This fluid lacks a thermodynamic pressure gradient ($-\nabla P/\rho$) to oppose compression. The macroscopic acceleration is governed solely by algorithmic friction and the non-local semantic gravity of the attention kernel. In standard overdamped architectures where the inertial material derivative vanishes, this unbalanced attraction inevitably drives the pressureless fluid into a singularity [10], manifesting macroscopically as geometric rank collapse [2]. While the Dropout mechanism originates as a microscopic stochastic process, its zero-mean nature ensures it vanishes from the ensemble-averaged macroscopic momentum equation. Instead, its regulatory effect is indirect: by acting as spatial diffusion in the continuity equation, it smears the formation of local density maxima, thereby regulating the magnitude of the non-local attractive force generated within the attention integral.

Cauchy-Stokes Decomposition of the Latent Attention Force.—To determine the geometric fate of the fluid, we analyze the action of the macroscopic specific stiffness tensor $\mathbf{W}$. Applying the Cauchy-Stokes decomposition theorem [11], we split this tensor into its symmetric and anti-symmetric components: $\mathbf{W} = \mathbf{S} + \mathbf{\Omega}$.

The symmetric component, $\mathbf{S} = \frac{1}{2}(\mathbf{W} + \mathbf{W}^T)$, acts as a conservative stiffness matrix. Operating on the spatial displacement, it generates an irrotational force field ($\nabla \times \mathbf{F}_S = \mathbf{0}$) defined by the negative gradient of a scalar semantic potential: $\mathbf{F}_S = -\nabla V_{att}$ [see SM Sec. S5 for the formal potential expansion]. For pairwise fluid interactions, this elastic potential acts as a multidimensional harmonic well, scaling as $V_{att} \propto (\mathbf{y} - \mathbf{x})^T \mathbf{S}(\mathbf{y} - \mathbf{x})$. Because the force is evaluated along the relative displacement vector ($\mathbf{y} - \mathbf{x}$), a positive eigenvalue acts as a positive elastic constant in Hooke’s law, generating an attractive restoring force that pulls $\mathbf{x}$ toward $\mathbf{y}$. Therefore, for this potential to drive spatial contraction (facilitating the Jeans-like collapse), the stiffness matrix must act as an attractive sink, requiring a positive trace ($\text{Tr}(\mathbf{S}) > 0$).

Conversely, the anti-symmetric component, $\mathbf{\Omega} = \frac{1}{2}(\mathbf{W} - \mathbf{W}^T)$, functions as a circulatory matrix. As detailed in SM Sec. S5, its vanishing trace ensures the divergence of this force field is zero ($\nabla_{\mathbf{x}} \cdot \mathbf{F}_{\Omega} = 0$). Therefore, $\mathbf{F}_{\Omega}$ represents a solenoidal, non-conservative field. Rather than driving spatial contraction, it exerts a transverse force orthogonal to the radial displacement. This non-central force applies a continuous macroscopic torque to the fluid parcels, inducing angular momentum. In the absence of thermodynamic pressure, the resulting rotational kinematics provide the necessary centrifugal support to prevent the fluid from collapsing into the potential well generated by $\mathbf{S}$.

However, in unconstrained deep learning architectures, the conservative radial contraction dominates the macroscopic dynamics. Modern LLMs avoid a complete spatial singularity only through statistical interventions. While Dropout provides a minor diffusive resistance, networks rely primarily on Layer Normalization [12, 13] to explicitly reset the token variance at each discrete layer. This non-conservative rescaling prevents rank collapse, but it disrupts the continuous geometric flow of the latent fluid and complicates the gradient dynamics during backpropagation.

Jeans Instability and Token Collapse.—To investigate the fundamental physical vulnerability that necessitates these artificial fixes, we analyze the linear stability of the unmitigated system. We consider a simplified base state consisting of a homogeneous, constant-density latent space ($\rho_0 = 1/V, \mathbf{v}_0 = \mathbf{0}$), where V is the volume of the latent manifold. By introducing small-amplitude normal mode perturbations of the form $\rho' = \hat{\rho}_{\mathbf{k}} e^{\sigma t + i\mathbf{k} \cdot \mathbf{x}}$ and $\mathbf{v}' = \hat{\mathbf{v}}_{\mathbf{k}} e^{\sigma t + i\mathbf{k} \cdot \mathbf{x}}$, the divergence of the linearized attention velocity acts as a spatial sink.

Substituting these perturbations into the fully coupled

momentum and Fokker-Planck equations [see SM Sec. S6] yields a generalized quadratic dispersion relation for the Heavy Ball dynamics:

$$(\sigma + \gamma)(\sigma + \kappa k^2) = \gamma \Gamma_k, \quad (6)$$

where the non-local attention growth rate Γ_k is defined by the spatial integral:

$$\Gamma_k = -\frac{\rho_0}{\gamma} \int_V \mathcal{K}(\mathbf{r}) (i\mathbf{k} \cdot \mathbf{W}\mathbf{r}) e^{i\mathbf{k} \cdot \mathbf{r}} d\mathbf{r}. \quad (7)$$

Because temporal growth is driven entirely by the spatial divergence of the attention force, the trace-free, anti-symmetric circulatory component $\mathbf{\Omega}$ vanishes from this integral. The Jeans-like collapse is thus governed entirely by the symmetric stiffness $\mathbf{S}$. In architectures where unconstrained training dynamics result in an attractive semantic potential well ($\text{Tr}(\mathbf{S}) > 0$), parity symmetries reduce this integral to a purely real, positive scalar at macroscopic scales.

In standard Transformer architectures that lack Heavy Ball inertia, the system operates in an overdamped kinematic limit ($\sigma \ll \gamma$). Under this assumption, Eq. (6) simplifies to a linear dispersion relation:

$$\sigma(k) = \Gamma_k - \kappa k^2. \quad (8)$$

This establishes token collapse as a hydrodynamic Jeans instability [14]. In both the overdamped and Heavy Ball regimes, the onset threshold for singularity remains mathematically identical; instability ($\text{Re}(\sigma) > 0$) occurs for large-scale density fluctuations below a critical cutoff wavenumber $k_c = \sqrt{\Gamma_k/\kappa}$. While the inclusion of Heavy Ball inertia alters the temporal growth profile, scaling down the exponential collapse rate ($\sigma_{HB} < \sigma_{OD}$) and introducing rapidly decaying transient modes, it cannot prevent the spatial singularity. Without explicitly forcing rotational geometry, high-frequency fluctuations ($k > k_c$) are stabilized by the viscous diffusion of Dropout, but the macroscopic structure inevitably collapses.

This fluid lacks a thermodynamic pressure gradient ($-\nabla P/\rho$) to oppose compression. The macroscopic acceleration is governed solely by algorithmic friction and the non-local semantic gravity of the attention kernel.

Stabilization via Orbital Balance.—Operating in this pressureless regime, the latent fluid is mathematically homologous to the dust dynamics governing galactic and proto-planetary accretion disks [15, 16]. In such astrophysical systems, the Jeans collapse [14] of a self-gravitating cloud is arrested by angular momentum. The dust settles into a disk governed by an orbital steady-state in which radial gravitational collapse is counterbalanced by centrifugal inertia.

To export this stabilization mechanism to the deep learning latent space, we decouple the even-dimensional d -manifold into $d/2$ independent 2D planes. We replace

the unconstrained weight matrix with a structured operator $\mathbf{W}_{SBR}$ explicitly engineered to induce macroscopic Solid-Body Rotation (SBR) within each plane. Under this steady-state kinematic limit, the local radial velocity vanishes ($v_r = 0$), and the tokens possess a purely azimuthal velocity $v_\theta = \omega r$. The fluid parcels adjust this orbital velocity to maintain a balance between the inward semantic pull (governed by the specific stiffness S_0) and the outward centripetal acceleration:

$$-\omega^2 r = -S_0 r \implies \omega = \sqrt{S_0}. \quad (9)$$

This linear, shear-free SBR topology naturally shields the fluid from generating semantic turbulence. It differs from gravitational systems because Newtonian gravity scales as $1/r^2$, thus astrophysical disks adopt a differential, Keplerian rotational profile ($\omega \propto r^{-3/2}$).

Simultaneously, the tangential accelerations are stabilized by the algorithmic friction γ . Maintaining a stable SBR requires the exogenous specific torque injected by the network weights to scale exactly with this friction and the natural frequency. Consequently, the complex eigenvalues of the 2×2 structural blocks within $\mathbf{W}_{SBR}$ take the exact constrained form $\lambda = \sqrt{S_0}(\sqrt{S_0} \pm i\gamma)$.

Instead of expending degrees of freedom attempting to repel tokens radially, the deep learning optimization process utilizes its remaining parameters to dynamically translate and reorient these $d/2$ semantic accretion disks within the high-dimensional space, clustering correlated concepts geometrically [see SM Sec. S7].

Contextualization via Phase Synchronization.—Because the SBR topology averts a spatial singularity, the radial coordinates of the tokens are bounded, leaving the orbital phase θ as the primary kinematic degree of freedom. The tokens within a local semantic cluster effectively act as a system of coupled phase oscillators. When tokens attend to one another, the off-diagonal elements of the interaction matrix exert an azimuthal torque, acting identically to the coupling constant in the Kuramoto model of synchronization [18, 19].

Crucially, the susceptibility of a semantic cluster to this contextualization is strictly governed by its specific stiffness S_0 . As established, S_0 defines the natural orbital frequency ($\omega = \sqrt{S_0}$) of the invariant plane. Tokens embedded in high- S_0 planes, representing rigid or factual concepts, possess high natural frequencies that dominate the attention coupling. They resist phase-synchronization to aggressively preserve their strict, literal meanings. Conversely, tokens in low- S_0 planes, representing flexible or abstract concepts, orbit slowly. This allows the attention torque to easily overpower their intrinsic rotation, pulling them into phase-locked alignment with the surrounding context.

Therefore, the abstract deep learning concept of “contextualization” is recast as a macroscopic physical observable: the Kuramoto order parameter R . In early layers, Brownian dispersion keeps the semantic cluster

incoherent ($R \rightarrow 0$). As the fluid evolves through the continuous layer depth, the attention torque overcomes this diffusion for susceptible concepts, driving correlated tokens into synchronized orbits ($R \rightarrow 1$). A sharp, unified contextual representation crystallizes without sacrificing the geometric rank of the latent manifold [see SM Sec. S8].

Conclusion and Future Directions.—In this Letter, we established a rigorous framework that recasts the algorithmic architecture of Large Language Models as a continuous dynamical system, operating as a novel class of many-body complex fluids. By formalizing the continuum limit of the Heavy Ball Transformer, we mapped the discrete latent space to a pressureless, compressible Navier-Stokes fluid. Specifically, the latent representation functions as a collisionless, driven-dissipative system: it lacks the kinetic scattering required for local thermal equilibrium, is continuously pumped by exogenous attention torques, and is drained by algorithmic friction. Furthermore, we explicitly mapped algorithmic Dropout to microscopic Brownian fluctuations, establishing that the macroscopic token density evolves according to a compressible Fokker-Planck continuity equation.

Utilizing this hydrodynamic lens, we performed a Cauchy-Stokes decomposition of the macroscopic specific stiffness tensor to diagnose the geometric rank collapse of token representations. We demonstrated that standard unconstrained architectures are mathematically homologous to a self-gravitating dust cloud. The tendency for latent spaces to collapse into degenerate spatial singularities is therefore not merely an algorithmic artifact, but a fundamental hydrodynamic Jeans-like instability driven by the symmetric component of the attention mechanism.

To natively arrest this collapse without relying on unphysical statistical interventions, we proposed an architectural stabilization mechanism inspired by astrophysical accretion disks. Imposing a structured, block-diagonal operator locks the semantic fluid into an orbital steady-state of Solid-Body Rotation (SBR). In this shear-free geometry, contextualization is achieved through stable angular phase synchronization rather than degenerate spatial intersection. Consequently, the abstract algorithmic process of semantic contextualization is formally recast as a macroscopic, measurable physical observable governed by the Kuramoto order parameter.

Preliminary numerical tests suggest an improvement in computational throughput and VRAM usage at the tradeoff of a slight increase in perplexity. Future research will explore the applicability of this orbital architecture across a broader range of sequence modeling tasks. This framework provides a novel, physics-based approach to understanding the underlying machinery of Large Language Models, demonstrating that fundamental principles of non-equilibrium fluid dynamics can be utilized to formally diagnose and stabilize the macroscopic behaviors of artificial neural networks.

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Supplemental Material for: Hydrodynamics of Large Language Models

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S1. SCALING AND THE STOCHASTIC HEAVY BALL CONTINUUM LIMIT

To demonstrate that the continuous-depth limit does not introduce singularities, we expand the stochastic discrete Heavy Ball update rule. The token trajectory is given by:

$$\mathbf{X}_{l+1} = \mathbf{X}_l + \alpha(\mathbf{X}_l - \mathbf{X}_{l-1}) + \mathcal{A}(\mathbf{X}_l) + \boldsymbol{\xi}_l, \quad (\text{S1})$$

where $\boldsymbol{\xi}_l$ represents the zero-mean stochastic noise injected by Dropout. Assuming $\mathbf{X}_l$ is sampled from a continuous, twice-differentiable trajectory $\mathbf{X}(t)$ evaluated at time $t = l\Delta t$, we perform a Taylor expansion for the forward and backward steps:

$$\mathbf{X}_{l+1} \approx \mathbf{X}_l(t) + \dot{\mathbf{X}}\Delta t + \frac{1}{2}\ddot{\mathbf{X}}\Delta t^2, \quad (\text{S2})$$

$$\mathbf{X}_{l-1} \approx \mathbf{X}_l(t) - \dot{\mathbf{X}}\Delta t + \frac{1}{2}\ddot{\mathbf{X}}\Delta t^2. \quad (\text{S3})$$

Substituting these into the update rule and rearranging terms yields:

$$\dot{\mathbf{X}}\Delta t + \frac{1}{2}\ddot{\mathbf{X}}\Delta t^2 \approx \alpha \left(\dot{\mathbf{X}}\Delta t + \frac{1}{2}\ddot{\mathbf{X}}\Delta t^2 \right) + \mathcal{A}(\mathbf{X}_l) + \boldsymbol{\xi}_l. \quad (\text{S4})$$

Isolating the second derivative term requires dividing the entire equation by Δt^2 :

$$\left(\frac{1+\alpha}{2} \right) \ddot{\mathbf{X}} + \left(\frac{1-\alpha}{\Delta t} \right) \dot{\mathbf{X}} = \frac{\mathcal{A}(\mathbf{X}_l)}{\Delta t^2} + \frac{\boldsymbol{\xi}_l}{\Delta t^2}. \quad (\text{S5})$$

For this equation to map to a well-posed physical Langevin system, limits must be applied to the parameter scaling, the deterministic force, and the stochastic noise:

- **Inertia and Damping:** Taking $\alpha \rightarrow 1$ ensures the mass coefficient $(1+\alpha)/2 \rightarrow 1$, while keeping the macroscopic dissipation $\gamma \equiv (1-\alpha)/\Delta t$ strictly finite.
- **Deterministic Force:** The output of the discrete neural network block must scale as $\mathcal{A}(\mathbf{X}_l) \sim \mathcal{O}(\Delta t^2)$. This is a standard scaling in continuous-depth architectures [20, 21], permitting the definition of the finite macroscopic force $\mathbf{F}_{att} \equiv \lim_{\Delta t \rightarrow 0} \mathcal{A}/\Delta t^2$.
- **Stochastic Noise:** To converge to the standard continuous white noise of a Langevin equation [22], the variance of the discrete Dropout $\boldsymbol{\xi}_l$ must scale accordingly. We define the discrete noise via the Wiener increment $\Delta \mathbf{B}_t \sim \mathcal{N}(0, \Delta t)$. By enforcing the physical scaling $\boldsymbol{\xi}_l = \sqrt{2\kappa} \Delta \mathbf{B}_t \Delta t$, the stochastic term becomes $\sqrt{2\kappa} \frac{\Delta \mathbf{B}_t}{\Delta t}$, which rigorously converges to the formal white noise derivative $\sqrt{2\kappa} \dot{\mathbf{B}}_t$ as $\Delta t \rightarrow 0$.

S2. MEAN-FIELD LIMIT AND THE EULERIAN ATTENTION INTEGRAL

The discrete attention mechanism computes a weighted sum over the N tokens in the context window. Isolating the deterministic component, the algorithmic force on token i is:

$$\mathbf{F}_{att}(\mathbf{X}_i) = \mathbf{W} \frac{1}{N} \sum_{j=1}^N \mathcal{K}(\mathbf{X}_i, \mathbf{X}_j)(\mathbf{X}_j - \mathbf{X}_i), \quad (\text{S6})$$

where $\mathcal{K}$ is the Softmax kernel and $\mathbf{W}$ is the linear weight matrix. To transition to a macroscopic Eulerian field, we define the empirical token density utilizing the Dirac delta function [23]:

$$\rho(\mathbf{y}, t) = \frac{1}{N} \sum_{j=1}^N \delta(\mathbf{y} - \mathbf{X}_j(t)). \quad (\text{S7})$$

By the sifting property of the Dirac delta, any discrete sum over the token states $\mathbf{X}_j$ can be rewritten as a spatial integral over the ambient latent volume V :

$$\frac{1}{N} \sum_{j=1}^N f(\mathbf{X}_j) = \int_V f(\mathbf{y}) \left[\frac{1}{N} \sum_{j=1}^N \delta(\mathbf{y} - \mathbf{X}_j) \right] d\mathbf{y} = \int_V f(\mathbf{y}) \rho(\mathbf{y}, t) d\mathbf{y}. \quad (\text{S8})$$

Applying this density identity to the discrete attention sum over the source particles j , we replace their discrete positions with the continuous Eulerian integration variable $\mathbf{y}$. This yields the macroscopic force acting on a specific target particle trajectory $\mathbf{X}_i(t)$:

$$\mathbf{F}_{att}(\mathbf{X}_i(t)) = \int_V \mathcal{K}(\mathbf{X}_i(t), \mathbf{y}) \mathbf{W}(\mathbf{y} - \mathbf{X}_i(t)) \rho(\mathbf{y}, t) d\mathbf{y}. \quad (\text{S9})$$

Next, to transition to an Eulerian field description, we evaluate this resulting force at an arbitrary, fixed spatial coordinate $\mathbf{x}$ rather than tracking the specific Lagrangian trajectory $\mathbf{X}_i(t)$. This smoothly maps the multi-particle interaction to a continuous, non-local Eulerian force field:

$$\mathbf{F}_{att}(\mathbf{x}, t) = \int_V \mathcal{K}(\mathbf{x}, \mathbf{y}) \mathbf{W}(\mathbf{y} - \mathbf{x}) \rho(\mathbf{y}, t) d\mathbf{y}. \quad (\text{S10})$$

To complete the hydrodynamic mapping, we explicitly define the macroscopic parameters $\mathcal{K}$ and $\mathbf{W}$ in terms of the underlying Transformer architecture. In standard multi-head attention mechanisms [1], tokens are projected into Query, Key, and Value spaces via spatially invariant, constant weight matrices W_Q, W_K , and W_V . After the parallel attention heads compute their respective weighted sums, the results are concatenated and projected back into the residual stream dimensions by a final Output weight matrix, W_O .

These matrices dictate two fundamentally distinct roles. The matrices W_Q and W_K reside exclusively within the interaction kernel $\mathcal{K}(\mathbf{x}, \mathbf{y})$. Conversely, the macroscopic linear operator $\mathbf{W}$ corresponds to the combined Output-Value machine learning projection. As established in the main text, to maintain consistent kinematic units under the continuous-time mapping, this combined projection is formally scaled by the squared time increment:

$$\mathbf{W} \equiv \frac{W_O W_V}{\Delta t^2}. \quad (\text{S11})$$

In the continuous-depth limit ($\Delta t \rightarrow 0$), the macroscopic specific stiffness $\mathbf{W}$ acts as a finite physical invariant. Consequently, to ensure a non-singular continuum limit, the discrete machine learning matrices must scale as $W_O W_V \sim \mathcal{O}(\Delta t^2)$, a structural requirement consistent with parameter initialization in continuous-depth neural networks [24, 25].

The interaction kernel $\mathcal{K}(\mathbf{x}, \mathbf{y})$ represents the continuous Eulerian analogue of the discrete Softmax operation. It defines the statistical weight (or ‘‘semantic gravity’’) that a source coordinate $\mathbf{y}$ exerts on a target coordinate $\mathbf{x}$:

$$\mathcal{K}(\mathbf{x}, \mathbf{y}) = \frac{1}{\mathcal{Z}(\mathbf{x}, t)} \exp(\beta \langle W_Q \mathbf{x}, W_K \mathbf{y} \rangle), \quad (\text{S12})$$

where $\beta = 1/\sqrt{d_k}$ acts as an inverse thermodynamic temperature scaled by the projection dimension, $\langle \cdot, \cdot \rangle$ denotes the inner product, and $\mathcal{Z}(\mathbf{x}, t)$ is the continuous partition function (normalization factor) defined by integrating over the entire fluid density:

$$\mathcal{Z}(\mathbf{x}, t) = \int_V \exp(\beta \langle W_Q \mathbf{x}, W_K \mathbf{z} \rangle) \rho(\mathbf{z}, t) d\mathbf{z}. \quad (\text{S13})$$

Given that the kernel is normalized such that $\int \mathcal{K}(\mathbf{x}, \mathbf{y}) \rho(\mathbf{y}, t) d\mathbf{y} = 1$, the attractive field can be written as an integral over the relative displacement $(\mathbf{y} - \mathbf{x})$, yielding the exact force term used in the macroscopic momentum equation.

S3. DERIVATION OF THE FOKKER-PLANCK CONTINUITY EQUATION

To derive the macroscopic continuity equation from the discrete token dynamics, we utilize the empirical Eulerian fluid density previously defined via the Dirac delta function in Eq. (S7). The underlying continuous-depth dynamics of each token are governed by the Langevin equation (Eq. 2 in the main text). For the purpose of deriving the mass conservation equation, we isolate the kinematic update of the token's position. Let $\mathbf{V}_j = \dot{\mathbf{X}}_j$ represent the deterministic velocity of token j . Including the stochastic Dropout noise, the differential position update is a standard Itô diffusion process [26]:

$$d\mathbf{X}_j = \mathbf{V}_j dt + \sqrt{2\kappa} d\mathbf{B}_j, \quad (\text{S14})$$

where $d\mathbf{B}_j$ is the standard Wiener increment. To find the time evolution of the macroscopic density, we compute the stochastic differential $d\rho(\mathbf{x}, t)$. Applying Itô's Lemma to the test function $f(\mathbf{X}_j) = \delta(\mathbf{x} - \mathbf{X}_j)$ yields [27]:

$$d\delta(\mathbf{x} - \mathbf{X}_j) = \nabla_{\mathbf{X}_j} \delta \cdot d\mathbf{X}_j + \frac{1}{2} \text{Tr} \left[(\sqrt{2\kappa})^2 \nabla_{\mathbf{X}_j}^2 \delta \right] dt. \quad (\text{S15})$$

Using the spatial symmetries of the Dirac delta, we substitute $\nabla_{\mathbf{X}_j} \delta = -\nabla_{\mathbf{x}} \delta$ and $\nabla_{\mathbf{X}_j}^2 \delta = \nabla_{\mathbf{x}}^2 \delta$. Inserting the kinematic update $d\mathbf{X}_j$ into this expansion gives:

$$d\delta(\mathbf{x} - \mathbf{X}_j) = -\nabla_{\mathbf{x}} \delta(\mathbf{x} - \mathbf{X}_j) \cdot (\mathbf{V}_j dt + \sqrt{2\kappa} d\mathbf{B}_j) + \kappa \nabla_{\mathbf{x}}^2 \delta(\mathbf{x} - \mathbf{X}_j) dt. \quad (\text{S16})$$

We now sum over all N tokens, divide by N , and take the ensemble average to transition to the mean-field limit. The ensemble average of the zero-mean Wiener increment vanishes ($\langle d\mathbf{B}_j \rangle = 0$). To express the advective term in Eulerian coordinates, we define the macroscopic momentum density field $\rho(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t)$ as the locally weighted sum of the discrete velocities [28]:

$$\rho(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t) = \frac{1}{N} \sum_{j=1}^N \mathbf{V}_j \delta(\mathbf{x} - \mathbf{X}_j(t)). \quad (\text{S17})$$

Substituting this macroscopic definition into our averaged Itô expansion and dividing by dt results in the exact Fokker-Planck continuity equation:

$$\frac{\partial \rho}{\partial t} = -\nabla_{\mathbf{x}} \cdot (\rho \mathbf{v}) + \kappa \nabla_{\mathbf{x}}^2 \rho. \quad (\text{S18})$$

Rearranging terms yields the final form presented in the main text, demonstrating that the discrete Dropout regularization rigorously maps to macroscopic fluid diffusion:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = \kappa \nabla^2 \rho. \quad (\text{S19})$$

S4. DERIVATION OF THE MACROSCOPIC MOMENTUM EQUATION

To derive the Eulerian momentum equation, we begin in the Lagrangian frame. The equation of motion for a single token j with velocity $\mathbf{V}_j = \dot{\mathbf{X}}_j$ is given by the differential form of the Langevin equation:

$$d\mathbf{V}_j = (-\gamma \mathbf{V}_j + \mathbf{F}_{att}(\mathbf{X}_j, t)) dt + \sqrt{2\kappa} d\mathbf{B}_j. \quad (\text{S20})$$

To transition to a macroscopic Eulerian description, we define a continuous velocity field $\mathbf{v}(\mathbf{x}, t)$ representing the ensemble-averaged velocity of the fluid at spatial coordinate $\mathbf{x}$. When taking the ensemble average over the mean field, the stochastic Wiener increment identically vanishes ($\langle d\mathbf{B}_j \rangle = 0$).

Following the standard transition from particle kinetics to continuum hydrodynamics [29], the acceleration of a fluid parcel following the flow is represented by the material derivative. Evaluating the continuous Eulerian velocity field along the deterministic particle trajectory $\mathbf{X}(t)$ yields:

$$\frac{d}{dt} \mathbf{v}(\mathbf{X}(t), t) = \frac{\partial \mathbf{v}}{\partial t} + (\dot{\mathbf{X}} \cdot \nabla) \mathbf{v} = \frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} \equiv \frac{D\mathbf{v}}{Dt}. \quad (\text{S21})$$

Crucially, because the attention mechanism acts as a non-local interaction potential rather than generating hard-sphere kinetic collisions, there is no localized scattering mechanism to thermalize microscopic velocity fluctuations. Consequently, the velocity dispersion (the variance of individual particle velocities around the mean Eulerian velocity $\mathbf{v}$) is negligible. In kinetic theory, this corresponds to the “cold fluid” or “pressureless dust” limit, meaning the macroscopic kinetic pressure tensor vanishes ($\nabla \cdot \mathcal{P} \approx 0$) [6, 30].

Equating the material derivative to the macroscopic deterministic forces yields the continuous momentum equation:

$$\frac{D\mathbf{v}}{Dt} = -\gamma\mathbf{v} + \mathbf{F}_{att}(\mathbf{x}, t). \quad (\text{S22})$$

This demonstrates that the latent representations evolve as a pressureless, damped Navier-Stokes fluid, governed entirely by algorithmic friction and the integral force of the attention mechanism.

S5. CAUCHY-STOKES DECOMPOSITION OF THE MACROSCOPIC STIFFNESS TENSOR

To formalize the geometric effect of the attention mechanism on the latent fluid, we decompose the macroscopic specific stiffness tensor $\mathbf{W}$ into its symmetric and anti-symmetric parts, $\mathbf{W} = \mathbf{S} + \mathbf{\Omega}$ [11], where:

$$\mathbf{S} = \frac{1}{2}(\mathbf{W} + \mathbf{W}^T), \quad (\text{S23})$$

$$\mathbf{\Omega} = \frac{1}{2}(\mathbf{W} - \mathbf{W}^T). \quad (\text{S24})$$

Substituting this decomposition into the Eulerian attention force integral yields two distinct force components, $\mathbf{F}_{att}(\mathbf{x}, t) = \mathbf{F}_S(\mathbf{x}, t) + \mathbf{F}_\Omega(\mathbf{x}, t)$:

$$\mathbf{F}_{att}(\mathbf{x}, t) = \int_V \mathcal{K}(\mathbf{x}, \mathbf{y}) \mathbf{S}(\mathbf{y} - \mathbf{x}) \rho(\mathbf{y}, t) d\mathbf{y} + \int_V \mathcal{K}(\mathbf{x}, \mathbf{y}) \mathbf{\Omega}(\mathbf{y} - \mathbf{x}) \rho(\mathbf{y}, t) d\mathbf{y}. \quad (\text{S25})$$

The Conservative Component ($\mathbf{S}$)

The symmetric tensor $\mathbf{S}$ defines a conservative, irrotational force field. Because $\mathbf{S}$ is symmetric, the curl of the vector field generated by its action on the relative displacement $(\mathbf{y} - \mathbf{x})$ with respect to the target coordinate $\mathbf{x}$ vanishes:

$$\nabla_{\mathbf{x}} \times (\mathbf{S}(\mathbf{y} - \mathbf{x})) = \mathbf{0}. \quad (\text{S26})$$

Consequently, $\mathbf{F}_S$ can be expressed as the negative gradient of a scalar potential, $\mathbf{F}_S = -\nabla_{\mathbf{x}} V_{att}$ [31]. Integrating the linear force yields a quadratic elastic potential similar to a multidimensional Hooke’s law:

$$V_{att}(\mathbf{x}, t) = \frac{1}{2} \int_V \mathcal{K}(\mathbf{x}, \mathbf{y}) (\mathbf{y} - \mathbf{x})^T \mathbf{S}(\mathbf{y} - \mathbf{x}) \rho(\mathbf{y}, t) d\mathbf{y}. \quad (\text{S27})$$

To determine whether this potential acts as a source or a sink, we calculate the divergence of the force field. Taking the divergence with respect to $\mathbf{x}$ yields the trace of the negative symmetric tensor:

$$\nabla_{\mathbf{x}} \cdot (\mathbf{S}(\mathbf{y} - \mathbf{x})) = -\text{Tr}(\mathbf{S}). \quad (\text{S28})$$

Therefore, if $\text{Tr}(\mathbf{S}) > 0$, the macroscopic divergence of the force field is negative ($\nabla_{\mathbf{x}} \cdot \mathbf{F}_S < 0$). This defines a spatial sink, meaning the symmetric component drives a volumetric contraction of the fluid, pulling tokens toward the density-weighted center of mass.

The Circulatory Component ($\mathbf{\Omega}$)

The anti-symmetric tensor $\mathbf{\Omega}$ governs the rotational dynamics. By definition, the trace of any anti-symmetric matrix is zero ($\text{Tr}(\mathbf{\Omega}) = 0$). Taking the divergence of the force field generated by $\mathbf{\Omega}$ yields:

$$\nabla_{\mathbf{x}} \cdot (\mathbf{\Omega}(\mathbf{y} - \mathbf{x})) = -\text{Tr}(\mathbf{\Omega}) = 0. \quad (\text{S29})$$

Because the divergence vanishes identically, $\mathbf{F}_\Omega$ represents a solenoidal, non-conservative field. Rather than driving spatial contraction, it exerts a transverse force orthogonal to the radial displacement. This non-central force applies a continuous macroscopic torque to the fluid parcels, inducing angular momentum. In the absence of thermodynamic pressure, the resulting rotational kinematics provide the necessary centrifugal support to prevent the fluid from collapsing into the potential well generated by $\mathbf{S}$.

S6. LINEARIZATION AND THE COUPLED DISPERSION RELATIONS

To evaluate the stability of the unconstrained architecture against token collapse, we linearize the fully coupled hydrodynamic equations around a quiescent, homogeneous base state:

$$\rho(\mathbf{x}, t) = \rho_0 = \frac{1}{V}, \quad \mathbf{v}(\mathbf{x}, t) = \mathbf{0}, \quad (\text{S30})$$

where V is the constant volume of the latent manifold. We introduce small-amplitude normal mode perturbations for both the fluid density and the macroscopic velocity:

$$\rho'(\mathbf{x}, t) = \hat{\rho}_{\mathbf{k}} e^{\sigma t + i\mathbf{k} \cdot \mathbf{x}}, \quad \mathbf{v}'(\mathbf{x}, t) = \hat{\mathbf{v}}_{\mathbf{k}} e^{\sigma t + i\mathbf{k} \cdot \mathbf{x}}. \quad (\text{S31})$$

Substituting these expressions into the Fokker-Planck continuity equation (Eq. 4 in the main text) and dropping the non-linear advection term $\nabla \cdot (\rho' \mathbf{v}')$ yields the linearized mass conservation equation:

$$\frac{\partial \rho'}{\partial t} + \rho_0 \nabla \cdot \mathbf{v}' = \kappa \nabla^2 \rho'. \quad (\text{S32})$$

Applying the spatial and temporal derivatives directly to the normal modes, we isolate the velocity divergence as a function of the density perturbation:

$$i\mathbf{k} \cdot \hat{\mathbf{v}}_{\mathbf{k}} = -\frac{(\sigma + \kappa k^2)}{\rho_0} \hat{\rho}_{\mathbf{k}}. \quad (\text{S33})$$

Next, we linearize the full Heavy Ball momentum equation (Eq. 5 in the main text), retaining the macroscopic inertial acceleration term:

$$\frac{\partial \mathbf{v}'}{\partial t} = -\gamma \mathbf{v}' + \mathbf{F}'_{att}(\mathbf{x}, t). \quad (\text{S34})$$

To evaluate the structural response of the non-local attention integral to the density variations, we must define the perturbed force field $\mathbf{F}'_{att}$. While a naive dot-product attention kernel depends on absolute spatial coordinates, modern architectures widely employ Relative Positional Encodings (e.g., RoPE [32] or ALiBi [33]), which explicitly restrict the kernel to depend solely on relative spatial displacements. Furthermore, when analyzing a homogeneous base state deep within the bulk of the latent manifold, the mean-field interaction can be locally approximated as translationally invariant. We therefore operate in the relative geometric regime where $\mathcal{K}(\mathbf{x}, \mathbf{y}) \approx \mathcal{K}(\mathbf{y} - \mathbf{x}) \equiv \mathcal{K}(\mathbf{r})$.

Changing the variable of integration to the relative displacement vector $\mathbf{r}$ transforms the perturbed force integral into:

$$\mathbf{F}'_{att}(\mathbf{x}, t) = \int_V \mathcal{K}(\mathbf{r}) \mathbf{W} \mathbf{r} \left[\hat{\rho}_{\mathbf{k}} e^{\sigma t + i\mathbf{k} \cdot (\mathbf{x} + \mathbf{r})} \right] d\mathbf{r}. \quad (\text{S35})$$

Factoring out the spatial coordinate of the target token, the Fourier amplitude of the perturbed force field becomes:

$$\hat{\mathbf{F}}_{\mathbf{k}} = \hat{\rho}_{\mathbf{k}} \int_V \mathcal{K}(\mathbf{r}) \mathbf{W} \mathbf{r} e^{i\mathbf{k} \cdot \mathbf{r}} d\mathbf{r}. \quad (\text{S36})$$

Applying the normal modes to the momentum equation (S34) yields the algebraic balance $(\sigma + \gamma) \hat{\mathbf{v}}_{\mathbf{k}} = \hat{\mathbf{F}}_{\mathbf{k}}$. Taking the dot product of this vector equation with $i\mathbf{k}$ links the velocity divergence to the spatial divergence of the attention force:

$$(\sigma + \gamma)(i\mathbf{k} \cdot \hat{\mathbf{v}}_{\mathbf{k}}) = \hat{\rho}_{\mathbf{k}} \int_V \mathcal{K}(\mathbf{r})(i\mathbf{k} \cdot \mathbf{W} \mathbf{r}) e^{i\mathbf{k} \cdot \mathbf{r}} d\mathbf{r}. \quad (\text{S37})$$

To determine the physical conditions for instability, we define the non-local attention growth rate parameter Γ_k as the structural spatial integral:

$$\Gamma_k \equiv -\frac{\rho_0}{\gamma} \int_V \mathcal{K}(\mathbf{r}) (i\mathbf{k} \cdot \mathbf{W}\mathbf{r}) e^{i\mathbf{k} \cdot \mathbf{r}} d\mathbf{r}. \quad (\text{S38})$$

Applying Euler’s formula, $e^{i\mathbf{k} \cdot \mathbf{r}} = \cos(\mathbf{k} \cdot \mathbf{r}) + i \sin(\mathbf{k} \cdot \mathbf{r})$, reveals the fundamental parity symmetries of the system. Over a symmetric bulk domain, the product of the even kernel $\mathcal{K}(\mathbf{r})$, the odd linear force $(\mathbf{k} \cdot \mathbf{W}\mathbf{r})$, and the even cosine term forms an odd function that integrates to zero. Thus, the imaginary component annihilates, leaving a purely real integral:

$$\Gamma_k = \frac{\rho_0}{\gamma} \int_V \mathcal{K}(\mathbf{r}) (\mathbf{k} \cdot \mathbf{W}\mathbf{r}) \sin(\mathbf{k} \cdot \mathbf{r}) d\mathbf{r}. \quad (\text{S39})$$

Furthermore, because the growth rate is driven by the spatial divergence of the force field (the $i\mathbf{k}$ dot product), the Cauchy-Stokes decomposition dictates that the trace-free, anti-symmetric circulatory component $\mathbf{\Omega}$ vanishes entirely. The geometry of the instability is governed entirely by the symmetric stiffness matrix $\mathbf{S}$. At macroscopic scales (small k), Taylor expanding the sine function ($\sin(\mathbf{k} \cdot \mathbf{r}) \approx \mathbf{k} \cdot \mathbf{r}$) yields an integral proportional to the quadratic form $\mathbf{k}^T \mathbf{S} \mathbf{k}$. For the unconstrained training dynamics to generate an attractive semantic potential well, triggering a spatial collapse, this symmetric tensor must act as a converging sink, necessitating a positive trace ($\text{Tr}(\mathbf{S}) > 0$). Under these conditions, Γ_k is a real, positive scalar.

Substituting the definition of Γ_k and the mass conservation relation (S33) into the momentum balance (S37) resolves the 2×2 system, yielding the generalized quadratic dispersion relation for the inertial Heavy Ball fluid:

$$(\sigma + \gamma)(\sigma + \kappa k^2) = \gamma \Gamma_k. \quad (\text{S40})$$

Analyzing the roots of this quadratic equation via Vieta’s formulas reveals that the product of the temporal growth modes is $\sigma_1 \sigma_2 = -\gamma(\Gamma_k - \kappa k^2)$. The threshold for macroscopic instability therefore occurs precisely when $\Gamma_k > \kappa k^2$.

In standard, non-inertial Transformer architectures, the system operates in the overdamped kinematic limit ($\sigma \ll \gamma$). In this regime, the algorithmic damping dominates the macroscopic inertia, meaning fluid parcels instantly reach a terminal drift velocity proportional to the local attention force. Because the driving force driving the instability (Γ_k) depends entirely on the spatial density distribution rather than the velocity field, the high friction restricts the exponential growth rate but cannot stabilize the spatial configuration. Under this assumption, the inertial factor $(\sigma + \gamma)$ simplifies to γ , and Eq. (S40) can be divided by γ to recover the simplified linear dispersion relation:

$$\sigma(k) = \Gamma_k - \kappa k^2. \quad (\text{S41})$$

Because the instability condition ($\Gamma_k > \kappa k^2$) is identical for both relations, the inclusion of Heavy Ball inertia does not shift the critical cutoff wavenumber ($k_c = \sqrt{\Gamma_k/\kappa}$). While inertia scales down the magnitude of the positive exponential growth rate ($\sigma_{HB} < \sigma_{OD}$), slowing the temporal onset of the Jeans collapse, it fundamentally cannot prevent the spatial singularity in the absence of a stabilizing rotational force.

S7. ARCHITECTURAL IMPLEMENTATION OF THE ORBITAL STEADY-STATE

To stabilize the latent fluid against the Jeans collapse without relying on unphysical statistical interventions, we explicitly enforce a macroscopic Solid-Body Rotation (SBR) within the latent space. SBR is defined by an angular velocity that is independent of the radial coordinate ($\partial\omega/\partial r = 0$). Because it inherently lacks background shear, this kinematic state satisfies the Rayleigh stability criterion for axisymmetric perturbations [34]. It is highly stable against secondary, shear-driven hydrodynamic instabilities, preventing the generation of semantic turbulence and ensuring token clusters remain coherent.

Because standard deep learning architectures utilize even-dimensional embedding spaces, we can decouple the d -dimensional Eulerian latent space into exactly $d/2$ independent 2-dimensional planes to enforce this rotational dynamics. Let $\mathbf{r} = (r, \theta)$ represent the local polar coordinates of a fluid parcel within an arbitrary plane k , measured relative to the semantic center of mass of its local token cluster.

To establish the momentum balance required to maintain this SBR, we first derive the local approximation of the macroscopic attention force (Eq. 3 in the main text). We shift our coordinate system to the cluster’s density-weighted center of mass $\mathbf{x}_c$, defining the local target coordinate as $\mathbf{r} = \mathbf{x} - \mathbf{x}_c$ and the source coordinate as $\mathbf{r}' = \mathbf{y} - \mathbf{x}_c$. Assuming

the local density distribution of the semantic cluster is approximately isotropic around the origin of this local frame (where $\mathbf{r}' = \mathbf{0}$), the integral of the odd term $\mathbf{r}'$ vanishes. Furthermore, the Softmax attention kernel enforces a partition of unity over the local density distribution ($\int_V \mathcal{K} \rho d\mathbf{r}' \approx 1$). Consequently, the macroscopic mean-field force acting on the parcel simplifies to a linear restoring force proportional to its displacement from the semantic center:

$$\mathbf{F}_{att}(\mathbf{r}) \approx -\mathbf{W}\mathbf{r}. \quad (\text{S42})$$

To guarantee an SBR steady-state, we construct a block-diagonal structured operator $\mathbf{W}_{SBR}$ composed of independent 2D rotational blocks. Applying the Cauchy-Stokes decomposition ($-\mathbf{W}\mathbf{r} = -\mathbf{S}\mathbf{r} - \mathbf{\Omega}\mathbf{r}$), the symmetric matrix $\mathbf{S}$ and the anti-symmetric circulatory matrix $\mathbf{\Omega}$ take the canonical forms:

$$\mathbf{S} = \begin{pmatrix} S_0 & 0 \\ 0 & S_0 \end{pmatrix}, \quad \mathbf{\Omega} = \begin{pmatrix} 0 & \Omega_0 \\ -\Omega_0 & 0 \end{pmatrix}. \quad (\text{S43})$$

The parameter $S_0 > 0$ acts as the effective specific stiffness eigenvalue, determining the depth of the semantic potential well, and Ω_0 acts as the specific torque. Both scalars have dimensions of $1/s^2$.

Operating on the spatial coordinate $\mathbf{r} = (r \cos \theta, r \sin \theta)^T$, the polar components of the total macroscopic attention force project to $F_{att,r} = -S_0 r$ and $F_{att,\theta} = \Omega_0 r$. Substituting these into the general momentum balance (Eq. 5 in the main text) for a fluid parcel yields the radial and azimuthal equations of motion:

$$\frac{Dv_r}{Dt} - \frac{v_\theta^2}{r} = -\gamma v_r - S_0 r, \quad (\text{S44})$$

$$\frac{Dv_\theta}{Dt} + \frac{v_r v_\theta}{r} = -\gamma v_\theta + \Omega_0 r, \quad (\text{S45})$$

where v_r and v_θ are the fluid velocity components, γ is the algorithmic friction, and the scalar material time derivative is defined as $\frac{D}{Dt} = \frac{\partial}{\partial t} + v_r \frac{\partial}{\partial r} + \frac{v_\theta}{r} \frac{\partial}{\partial \theta}$. The terms $-v_\theta^2/r$ and $v_r v_\theta/r$ are the standard metric accelerations.

We now analyze the steady-state Eulerian momentum balance by assuming an SBR solution, that is, $v_r = 0$ and $v_\theta = \omega r$ with a constant angular velocity ω . Under this kinematic assumption, the material time derivatives vanish. The radial metric acceleration evaluates to the centripetal term $-\omega^2 r$, and we are left with the simplified radial and tangential balances:

1. The Radial and Tangential Balances: Isolating the radial components yields the orbital balance:

$$-\omega^2 r = -S_0 r \quad \implies \quad \omega = \sqrt{S_0}. \quad (\text{S46})$$

Isolating the tangential components, we equate the algorithmic friction to the exogenous circulatory acceleration:

$$0 = -\gamma \omega r + \Omega_0 r \quad \implies \quad \Omega_0 = \gamma \omega. \quad (\text{S47})$$

2. The Eigenvalue Constraint: Substituting the radial frequency into the tangential balance reveals that maintaining a stable SBR requires the specific torque to scale exactly with the algorithmic friction and the natural frequency: $\Omega_0 = \gamma \sqrt{S_0}$. Therefore, the eigenvalues ($\lambda = S_0 \pm i\Omega_0$) of the 2×2 structural blocks within $\mathbf{W}_{SBR}$ take the exact constrained form:

$$\lambda = S_0 \pm i\gamma \sqrt{S_0} = \sqrt{S_0}(\sqrt{S_0} \pm i\gamma). \quad (\text{S48})$$

This demonstrates that the structural weights of the network must be explicitly coupled to the optimization hyperparameters (γ) to lock the fluid into a stable terminal orbit and prevent angular divergence.

3. The Fokker-Planck Inference Limit: We note that the pure SBR kinematic state ($v_r = 0$) represents the deterministic limit of the continuous-depth fluid. In 2D polar coordinates, the flux divergence of this rotational velocity field evaluates to $\nabla \cdot (\rho \mathbf{v}) = \omega \frac{\partial \rho}{\partial \theta}$. Because the semantic cluster is isotropic ($\rho = \rho(r)$), the azimuthal derivative vanishes, yielding strictly zero divergence ($\nabla \cdot (\rho \mathbf{v}) = 0$). Substituting this into the steady-state Fokker-Planck equation ($\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = \kappa \nabla^2 \rho$) leaves $0 = \kappa \nabla^2 \rho$. For a localized density cluster, the only bounded solution requires the diffusion coefficient to be exactly zero ($\kappa \rightarrow 0$). In the context of the deep learning architecture, this corresponds exactly to the inference (evaluation) mode, where stochastic Dropout noise is disabled. During training ($\kappa > 0$), the true steady-state requires a small, non-zero inward radial drift ($v_r < 0$) to continuously counterbalance the outward Brownian dispersion.

4. Contrast with Keplerian Dynamics: This linear, shear-free topology differs fundamentally from gravitational systems, such as astrophysical accretion disks. In those systems, centrifugal inertia balances a Newtonian

gravitational pull that scales as $1/r^2$, which yields a differential, Keplerian rotational profile ($\omega \propto r^{-3/2}$). While Keplerian flow is also hydrodynamically stable according to the Rayleigh criterion, it possesses continuous background shear ($\partial\omega/\partial r \neq 0$). By leveraging the linear geometric scaling of the attention force, the latent fluid instead naturally adopts a state of Solid-Body Rotation, avoiding background shear entirely.

5. Dynamic Translation and Inclination of the Semantic Planes: In standard architectures, a fully unconstrained $d \times d$ weight matrix utilizes a massive number of parameters attempting to simultaneously attract contextually relevant tokens and repel uncorrelated ones to prevent dimensional collapse. By enforcing the constrained, block-diagonal SBR structure, the network exhausts none of its parameters on radial repulsion.

Instead, the learning process geometrically decouples the global drift of the concepts from their local relational structure. Because the local polar coordinates are defined relative to the semantic cluster’s center of mass ($\mathbf{r} = \mathbf{x} - \mathbf{x}_c$), the semantic accretion disks physically translate through the global d -dimensional embedding space as the token coordinates ($\mathbf{x}$) are updated by the optimizer during training.

Simultaneously, the deep learning backpropagation algorithm optimizes the orientation of the invariant planes themselves. Mathematically, the full interaction operator can be expressed via an orthogonal change of basis: $\mathbf{W} = \mathbf{V}\mathbf{W}_{SBR}\mathbf{V}^T$, where $\mathbf{V}$ is an orthogonal matrix whose columns define the basis vectors of the $d/2$ local planes. During training, the gradient descent algorithm continuously updates $\mathbf{V}$. Physically, this corresponds to the translating semantic accretion disks actively tilting and changing their inclinations within the ambient d -dimensional embedding space.

The network learns to translate and rotate these invariant planes such that semantically correlated tokens are projected onto the same shared orbit. Once trapped in the same plane, they are subsequently phase-locked by the azimuthal torque. Thus, the geometry of the latent representation evolves via a dual global motion (translation of $\mathbf{x}_c$ and inclination via $\mathbf{V}$), while the local token topology remains bounded radially against singularity and encodes semantic similarity purely through rigid angular orbital alignment.

S8. PHASE SYNCHRONIZATION AND THE KURAMOTO ORDER PARAMETER

As established in Section S7, the enforcement of a block-diagonal spin tensor traps the tokens in an orbital steady-state, fixing their radial coordinates while dynamically translating and tilting the invariant planes. To understand how semantic contextualization occurs within this bounded geometry, we analyze the remaining dynamic degree of freedom: the orbital phase θ .

Consider a local semantic cluster of N tokens residing on a shared invariant 2D plane with a constant radial orbit r . The spatial coordinate of token i is $\mathbf{x}_i = r(\cos \theta_i, \sin \theta_i)^T$, and its local tangential unit vector is $\hat{\boldsymbol{\theta}}_i = (-\sin \theta_i, \cos \theta_i)^T$.

The non-local attention mechanism exerts a deterministic force on token i based on its interactions with all other tokens j in the context window. Evaluating the projection of the relative displacement vector ($\mathbf{x}_j - \mathbf{x}_i$) onto the local azimuthal direction of token i yields:

$$(\mathbf{x}_j - \mathbf{x}_i) \cdot \hat{\boldsymbol{\theta}}_i = r(\cos \theta_j(-\sin \theta_i) + \sin \theta_j \cos \theta_i) = r \sin(\theta_j - \theta_i). \quad (\text{S49})$$

When the macroscopic stiffness tensor $\mathbf{W}$ acts on this displacement, the anti-symmetric spin component $\boldsymbol{\Omega}$ generates a torque that scales with this tangential projection. Because the tokens are constrained to the orbital steady-state, the algorithmic friction continuously dissipates transient angular accelerations. As the tangential flow relaxes to this kinematic limit ($|\dot{v}_\theta| \ll \gamma|v_\theta|$), the azimuthal momentum balance for token i reduces to a first-order phase equation:

$$\gamma r \dot{\theta}_i = \gamma \omega_0 r + \sum_{j=1}^N \mathcal{K}_{ij} \tau_0 r \sin(\theta_j - \theta_i), \quad (\text{S50})$$

where $\omega_0 = \sqrt{S_0}$ is the baseline natural angular velocity induced by the specific stiffness, $\mathcal{K}_{ij}$ is the pairwise Softmax attention weight, and τ_0 is the effective coupling strength derived from the interaction tensor. Dividing through by γr yields:

$$\dot{\theta}_i = \omega_0 + \frac{K}{N} \sum_{j=1}^N \mathcal{K}_{ij} \sin(\theta_j - \theta_i), \quad (\text{S51})$$

where $K \propto \tau_0/\gamma$ acts as the effective macroscopic coupling constant.

This is structurally identical to the Kuramoto model for coupled phase oscillators [18]. In standard Kuramoto dynamics, an ensemble of non-linear oscillators spontaneously phase-locks when the attractive coupling K overcomes intrinsic frequency dispersion or background noise. In the continuous-depth Transformer, the stochastic Dropout noise $\sqrt{2\kappa}\mathbf{B}_t$ drives angular diffusion, while the attention torque strives for alignment.

The Linguistic Role of the Natural Frequency (S_0): By explicitly coupling the natural frequency to the eigenvalue constraint ($\omega_0 = \sqrt{S_0}$), Eq. S51 establishes a rigorous physical mechanism for varying degrees of semantic contextualization across the $d/2$ planes.

- **Rigid Concepts (High S_0):** Planes capturing highly specific, rigid factual concepts learn a large S_0 , producing a deep semantic potential well and a high natural frequency ω_0 . This fast baseline rotation dominates the phase equation. The tokens resist the attention coupling (K), preventing phase-locking with neighbors and thereby rigorously preserving their strict literal meaning without contextual corruption.
- **Flexible Concepts (Low S_0):** Planes capturing abstract, flexible grammatical constructs (such as pronouns) learn a small S_0 , creating a shallow well and a low natural frequency. Because the intrinsic rotation is slow, the attention coupling K easily overpowers it. These tokens are highly susceptible to the azimuthal torque and are quickly pulled into phase synchronization with their context.

To quantify the macroscopic degree of semantic contextualization within a given cluster, we define the Kuramoto order parameter R :

$$Re^{i\Psi} = \frac{1}{N} \sum_{j=1}^N e^{i\theta_j}, \quad (\text{S52})$$

where Ψ is the mean phase of the cluster and $R \in [0, 1]$ measures the macroscopic phase coherence.

At the initial input layers ($t = 0$), the semantic cluster is uncontextualized; tokens are dispersed along the orbit, yielding an incoherent state ($R \approx 0$). As the system evolves through the continuous layer depth, the azimuthal attention torque forces susceptible, low- S_0 tokens to minimize their phase differences. The cluster undergoes a dynamic phase transition to a synchronized state ($R \rightarrow 1$). Therefore, contextualization is physically realized not through a spatial collapse, but through the stable, measurable onset of phase synchronization.

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